Chunqiu Steven Xia

Curriculum Vitae

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Research Interests

My research lies at the intersection of **Software Engineering** and **Artificial Intelligence**, with a central focus on synergizing fundamental software engineering principles with advanced AI technologies, particularly large language models (LLMs) and agentic systems. I am passionate about building rigorous foundations and scalable solutions to: 1) automate challenging and cognitively demanding software tasks, 2) enhance the reliability and trustworthiness of critical software systems, and 3) push the boundaries of AI coding and general reasoning through massive software-centric training, evaluation, and real-world applications.

To date, my research work has accumulated over **6,000** citations on Google Scholar and over **4,000** GitHub stars. Moreover, my recent work on LLM-based program synthesis has been adopted by leading AI companies (including OpenAI, Meta, and DeepSeek), with over one **million** downloads worldwide.

My future research vision is to revolutionize real-world software development by building autonomous LLM agents equipped with specialized software engineering tools and the ability to self-improve on the fly to solve complex and challenging tasks. I aim to not only enhance developer productivity and software reliability but also leverage software itself to create rich training data and environments that can improve reasoning and planning capabilities for the next generation of LLMs and agents

Education

2021 – Now (Expected July'26) **Ph.D. Computer Science**, *University of Illinois Urbana-Champaign (UIUC)*
Advisor: Lingming Zhang

2017 – 2021 **B.ASc in Computer Engineering**, *University of Toronto (UofT)*

Selected Honors & Awards

2025 ACM SIGSOFT Distinguished Paper at FSE 2025

2025 CACM Research Highlight for Repilot [9]

2024 – 2025 Amazon AICE Fellowship, *$60,000*

2024 Oral Paper Spotlight at COLM 2025

2020 UofT Electrical and Computer Engineering Top Student

2020 Adel S. Sedra Award In Electrical And Computer Engineering

2019 – 2020 Walberg Undergraduate Scholarship

2019 – 2021 Andrew Alexander Kinghorn Scholarship

2019 NSERC Undergraduate Student Research Award

2025 NSF Student Travel Award for FSE 2025

2022 – 2025 SIGSOFT CAPS Travel Award (FSE '25, ISSTA '24, ICSE '24/'23, FSE '22)

Publications & Preprints

[First/Co-First author papers] 10 (8 Top-tier with 1 under submission)

[Co-author papers] 7 (All Top-tier)

[Google Scholar Citations] Total #: 6,054; h-index: 15; i10-index: 16

* *indicates equal contribution*

Preprint [1] **Chunqiu Steven Xia**, Zhe Wang, Yan Yang, Yuxiang Wei, Lingming Zhang. Live-SWE-agent: Can Software Engineering Agents Self-Evolve on the Fly?, November 2025.
First *live* software agent that continuously evolve itself *on-the-fly*; **Top-1** open-source agent on SWE-bench Verified (**79.2%**) and SWE-Bench Pro (**45.8%**)

EMNLP '25 [2] Cheryl Lee, **Chunqiu Steven Xia**, Longji Yang, Jen-tse Huang, Zhouruixing Zhu, Lingming Zhang, Michael R. Ly. UniDebugger: Hierarchical Multi-Agent Framework for Unified Software Debugging. *In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*, August 2025.

FSE '25 [3] **Chunqiu Steven Xia***, Yinlin Deng*, Soren Dunn, Lingming Zhang. Demystifying LLM-Based Software Engineering Agents. *In Proceedings of 33rd ACM SIGSOFT International Symposium on the Foundations of Software Engineering (FSE 2025)*, June 2025.
⏳ ACM SIGSOFT Distinguished Paper Award; **1.9K** stars on GitHub; Used by OpenAI, Meta, DeepSeek, etc.

NeurIPS '24 [4] Yinlin Deng, **Chunqiu Steven Xia**, Zhezhen Cao, Meiziniu Li, Lingming Zhang. Can LLMs Implicitly Learn Numeric Parameter Constraints in Data Science APIs?. *In Proceedings of the 38th Annual Conference on Neural Information Processing Systems (NeurIPS 2024)*, December 2024.

COLM '24 [5] **Chunqiu Steven Xia***, Yinlin Deng*, Lingming Zhang. Top Leaderboard Ranking = Top Coding Proficiency, Always? EvoEval: Evolving Coding Benchmarks via LLM. *In Proceedings of the 1st Conference on Language Modeling (COLM 2024)*, October 2024.
⏳ Oral Paper Spotlight

ISSTA '24 [6] **Chunqiu Steven Xia** and Lingming Zhang. Keep the Conversation Going: Fixing 162 out of 337 bugs for $0.42 each using ChatGPT. *In Proceedings of the 33rd International Symposium on Software Testing and Analysis (ISSTA 2024)*[1], September 2024.
Inspired work on iterative repair, code, and test generation powered by LLMs; Concepts in this work has been widely adopted by modern software agents

ICSE '24 [7] **Chunqiu Steven Xia**, Matteo Paltenghi, Jia Le Tian, Michael Pradel, Lingming Zhang. Fuzz4All: Universal Fuzzing with Large Language Models. *In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (ICSE 2024)*, April 2024.
Found **98** bugs across **9** different popular software (e.g., C/C++/Go/Java compilers, SMT solvers, and quantum computing platform)

[1]The arXiv version of ChatRepair [6] was posted in January 2023

ICSE '24 [8] Yinlin Deng, **Chunqiu Steven Xia**, Chenyuan Yang, Shizhuo Dylan Zhang, Shujing Yang, Lingming Zhang. Large Language Models are Edge-Case Fuzzers: Testing Deep Learning Libraries via FuzzGPT. *In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (ICSE 2024)*, April 2024.
Found **76** bugs in Machine Learning systems (PyTorch and TensorFlow)

FSE '23 [9] Yuxiang Wei, **Chunqiu Steven Xia**, Lingming Zhang. Copiloting the Copilots: Fusing Large Language Models with Completion Engines for Automated Program Repair. *In Proceedings of the 31th ACM SIGSOFT International Symposium on the Foundations of Software Engineering (FSE 2023)*, December 2023.
⏳ CACM Research Highlight

NeurIPS '23 [10] Jiawei Liu*, **Chunqiu Steven Xia***, Yuyao Wang, Lingming Zhang. Is Your Code Generated by ChatGPT Really Correct? Rigorous Evaluation of Large Language Models for Code Generation. *In Proceedings of the 37th Annual Conference on Neural Information Processing Systems (NeurIPS 2023)*, December 2023.
1 million HuggingFace dataset downloads; Integrated by OpenAI, Amazon, DeepSeek, Google, Meta, etc.

ASE '23 [11] **Chunqiu Steven Xia**, Yifeng Ding, Lingming Zhang. The Plastic Surgery Hypothesis in the Era of Large Language Models. *In Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE 2023)*, September 2023.

ISSTA '23 [12] Yinlin Deng, **Chunqiu Steven Xia**, Haoran Peng, Chenyuan Yang, Lingming Zhang. Large Language Models are Zero-Shot Fuzzers: Fuzzing Deep-Learning Libraries via Large Language Models. *In Proceedings of the 32nd International Symposium on Software Testing and Analysis (ISSTA 2023)*, July 2023.
Opened up the entire new direction of LLM-based fuzzing; Found **65** bugs in Machine Learning systems (PyTorch and TensorFlow)

ICSE '23 [13] **Chunqiu Steven Xia**, Yuxiang Wei, Lingming Zhang. Automated Program Repair in the Era of Large Pre-trained Language Models. *In Proceedings of the 45th IEEE/ACM International Conference on Software Engineering (ICSE 2023)*, May 2023.

ICSE '23 [14] **Chunqiu Steven Xia**, Saikat Dutta, Sasa Misailovic, Darko Marinov, Lingming Zhang. Balancing Effectiveness and Flakiness of Non-Deterministic Machine Learning Tests. *In Proceedings of the 45th IEEE/ACM International Conference on Software Engineering (ICSE 2023)*, May 2023.

FSE '22 [15] **Chunqiu Steven Xia** and Lingming Zhang. Less Training, More Repairing Please: Revisiting Automated Program Repair via Zero-shot Learning. *In Proceedings of the 30th ACM SIGSOFT International Symposium on the Foundations of Software Engineering (FSE 2022)*, November 2022.
Demonstrated for the first time LLMs can outperform all prior APR techniques studied for over a decade; Pioneered the new direction of LLM-based APR

USENIX '21 [16] Weicheng Cao, **Chunqiu Steven Xia**, Sai Teja Peddinti, David Lie, Nina Taft, Lisa M. Austin. A Large Scale Study of User Behavior, Expectations and Engagement with Android Permissions. *In Proceedings of the 30th USENIX Security Symposium (USENIX 2021)*, August 2021.

MICRO '18 [17] Joshua San Miguel, Karthik Ganesan, Mario Badr, **Chunqiu Steven Xia**, Rose Li, Hsuan Hsiao, Natalie Enright Jerger. The eh model: Early design space exploration of intermittent processor architectures. *In Proceedings of the 51st Annual IEEE/ACM International Symposium on Microarchitecture (MICRO 2018)*, October 2018.

Selected Open-Source Software Contributions

my GitHub ID `https://github.com/brutalsavage`

Agentless >1.9K stars Agentless [3] is an *agentless* approach to automatically resolve software development issues. Agentless operates through well-defined stages (such as issue localization, patching, and test generation) inspired by the software literature, without relying on the LLM agent to autonomously decide future actions or use complex tools. Agentless has been adopted by top AI companies (e.g., OpenAI, Meta, and DeepSeek) to evaluate their recent models on real-world coding tasks and build their own custom issue-resolution scaffolds. Agentless is at `https://github.com/OpenAutoCoder/Agentless`.

Fuzz4All >300 stars Fuzz4All [7] is the first universal fuzzer using LLMs that can generate arbitrary programming languages and target a wide-range of systems under test. Fuzz4All leverages LLMs as both an input generator and also a mutation engine to produce diverse and realistic inputs for any target language. To accomplish this, Fuzz4All employs a novel autoprompting technique to synthesize tailored LLM prompts for each target and the LLM-based fuzzing loop to generate fuzzing inputs. We evaluated Fuzz4All on nine different systems (GCC, Clang, CVC5, Z3, Go, Java compilers and Qiskit quantum library) and uncovered 98 previously undiscovered bugs. Fuzz4all is at `https://github.com/fuzz4all/fuzz4all`

EvalPlus >1.6K stars EvalPlus [10] combines both LLM and mutation-based test generation to automatically generate vast amount of high-quality test cases for existing benchmarks. Surprisingly, we found that the performance of all LLMs on the new benchmark is much lower, showing for the first time that prior widely-used benchmarks contain insufficient test cases and do not accurately reflect the true performance of LLMs for code synthesis. The resulting augmented benchmarks have become the default benchmark to evaluate LLM-based program synthesis, used by hundreds of papers as well as adopted by multiple big-tech companies (e.g, OpenAI, Google, Meta, and IBM). EvalPlus is at `https://github.com/evalplus/evalplus`.

Service to Professional Community

PC Member International Conference on Automated Software Engineering (ASE 2025)

PC Member International Conference on AI foundation Models and Software Engineering (FORGE 2025)

PC Member International Workshop on Automated Program Repair (APR 2025)

PC Member International Workshop on Automated Program Repair (APR 2024)

Conference Reviewer ICLR 2026, NeurIPS 2025, COLM 2025, ICLR 2025, NeurIPS 2024

Journal Reviewer TSE 2025, TSE 2024, TOSEM 2024

Presentations and Invited Talks

Invited Talk *Demystifying LLM-Based Software Engineering Agents*, Meta, December 2025

Conference Talk	*Demystifying LLM-Based Software Engineering Agents*, FSE 2025, June 2025
Poster Presentation	*Can LLMs Implicitly Learn Numeric Parameter Constraints in Data Science APIs?*, NeurIPS 2024, December 2024
Conference Talk	*Top Leaderboard Ranking = Top Coding Proficiency, Always? EvoEval: Evolving Coding Benchmarks via LLM*, COLM 2024, October, 2024
Conference Talk	*Keep the Conversation Going: Fixing 162 out of 337 bugs for $0.42 each using ChatGPT* , ISSTA 2024, September 2024
Invited Talk	*Demystifying LLM-based Software Engineering Agents*, South Park Commons AI Agent Speaker Series, August 2024
Invited Talk	*Agents & LLMs for Software Development*, Alibaba Tongyu Lingma Team, July 2024
Conference Talk	*Universal Fuzzing with Large Language Models*, ICSE 2024, April 2024
Invited Talk	*Universal Fuzzing with Large Language Models*, SecurityFrontiers.AI, April 2024
Poster Presentation	*Is Your Code Generated by ChatGPT Really Correct? Rigorous Evaluation of Large Language Models for Code Generation*, NeurIPS 2023, December 2023
Invited Talk	*Universal Fuzzing with Large Language Models*, Kwai Inc, November 2023
Guest Lecture	*LLMs for Program Repair, Fuzzing and Testing*, NC State University, October 2023
Conference Talk	*The Plastic Surgery Hypothesis in the Era of Large Language Models*, ASE 2023, September 2023
Guest Lecture	*Are Large Language Models Effective Fuzzers?*, Ohio State University, April 2023
Conference Talk	*Automated Program Repair in the Era of Large Pre-trained Language Models*, ICSE 2023, May 2023
Conference Talk	*Balancing Effectiveness and Flakiness of Non-Deterministic Machine Learning Tests*, ICSE 2023, May 2023
Invited Talk	*Leveraging Large Language Models for Automated Program Repair ... and other Software Engineering tasks*, Uber Programming Systems Team, April 2023
Invited Talk	*Leveraging Large Language Models for Software Engineering Tasks*, Microsoft Data&AI, January 2023
Invited Talk	*Automated Program Repair in the era of Large Language Models*, Software Lab at the University of Stuttgart, December 2022
Conference Talk	*Less Training, More Repairing Please: Revisiting Automated Program Repair via Zero-shot Learning*, FSE 2022, November 2022

Students Mentored

Yifeng Ding (PhD, UIUC); co-authored [11]

Yuxiang Wei (PhD, UIUC); co-authored [1,9,13]

Yuyao Wang (Undergrad, Nanjing University → PhD, University Of Washington); co-authored [10]

Haoran Peng (Undergrad, University of Science and Technology of China → PhD, University Of Washington); co-authored [12]

Tom Tian (MS, UIUC → Databricks); co-authored [7]

Katherine Wang (MS, UIUC → Uber)

Zhezhen Cao (Undergrad, Southern University of Science and Technology → PhD, North Carolina State University); co-authored [4]

Soren Dunn (MS, UIUC → Intology AI); co-authored [3]

Teaching Experiences

UIUC, Spring 2024 **Teaching Assistant**, CS598: Software Quality Assurance with Generative AI
Top 1% TA in the department according to the UIUC ICES Course Evaluation for Spring 2024